

Jonathan Nguyen

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EDUCATION

University of California, Berkeley

Berkeley, CA

Bachelor of Arts in Applied Mathematics | Cumulative Transfer GPA: 3.93/4.00 Aug. 2026 – May 2028 (Expected)

- **Academic Focus:** Machine Learning, Optimization & Data Science
- **Planned Coursework:** CS 61A, CS 61B, CS 70, EECS 126, EECS 127, MATH 156, MATH 170, IND ENG 142A, IND ENG 142B

EXPERIENCE

SkyVault AI

Palo Alto, CA

AI Engineer Intern

Oct. 2025 – Present

- Engineered *SkyvaultQuarterback*, a Python and Claude platform that centralizes and manages a multibillion-dollar data-center sales pipeline in PostgreSQL/Supabase while preserving Excel import/export workflows
- Built a schema-constrained LLM extraction pipeline that converts emails, Read.ai transcripts, and screenshots into structured deal facts using Claude, Pydantic validation, source provenance, and model-call logging
- Implemented authenticated CLI workflows, PostgreSQL row-level security, audited writes, append-only change history, and reversible updates to secure and trace pipeline-data changes

Florida State University, Department of Mathematics

Tallahassee, FL

Undergraduate Research Assistant | Advisor: Dr. Astrit Tola

Sep. 2025 – Present

- Conducted machine-learning research on classifying autism (ABIDE), Alzheimer's, and Parkinson's disease from fMRI-derived functional-connectivity graphs using GNNs and Topological Data Analysis
- Helped improve ABIDE autism-classification accuracy from 58% to 71% (+13 percentage points) by iterating on graph-based deep-learning architectures in PyTorch
- Implemented an SPD-manifold learning pipeline with Fréchet mean estimation, tangent-space projection, and differentiable top- k graph sparsification via Gumbel-Softmax
- Designed an edges-as-nodes directed line-graph model for seven-label protein-protein interaction classification, connecting interaction nodes that share a protein endpoint
- Implemented five-view GraphSAGE and Polynormer branches with learned fusion across topology, ESM2 sequence embeddings, and pLDDT confidence features; added duplicate handling and split-overlap checks

Project Motion

Palo Alto, CA

Founder & Lead Developer

2024 – Present

- Built a computer-vision biomechanics platform using YOLOv8-L Pose to extract 17 body keypoints per video frame and reconstruct baseball swing kinematics; adopted by athletes at Stanford, Cal Poly, Brown, and Holy Cross
- Integrated pose estimates with a Unity/C# skeletal renderer and joint-kinematics pipeline to convert video into 3D motion, quantitative swing metrics, and coach-facing feedback

PROJECTS

AIDJ | Python, librosa

- Built an end-to-end music-information-retrieval and DSP system that estimates beat grids and musical key, orders tracks by tempo and Camelot compatibility, and renders phrase-aligned, beatmatched mixes

Black-Hole GPU Ray Tracer | C++, GLSL, OpenGL

- Built a GPU-accelerated ray tracer for Schwarzschild gravitational lensing, numerically integrating null geodesics in an OpenGL compute shader

TECHNICAL SKILLS

Languages: Python, C++, SQL, C#, Java, GLSL

ML & Data: PyTorch, PyTorch Geometric, scikit-learn, Ultralytics YOLO, OpenCV, NumPy, pandas, librosa

Models & Methods: Graph Neural Networks (GCN, GraphSAGE), Transformers, Attention, LLM Extraction, Computer Vision, Pose Estimation, Topological Data Analysis, SPD Manifold Learning

Systems & Tools: Claude API, Pydantic, PostgreSQL, Supabase, Docker, Kubernetes, Git, Unity, OpenGL